

BECM 3107

PROJECT COMMISSIONING & CLOSE-OUT

TERM FINAL STUDY GUIDE

Easy English + প্রয়োজনীয় বাংলা অর্থ + **Memory Techniques**

Based strictly on the uploaded 39-page lecture handout

WHAT THIS GUIDE CONTAINS

Commissioning meaning, agent, quality-assurance process and bid-document requirements
Project closeout process, staffing release, financial closure, lessons learned and archive
Contractor closing plan, customer sign-off, closeout checklist and report formats
Bangla vocabulary, memory codes, model answers, MCQs, scenarios and revision plan

Prepared for fast understanding, active recall and written-exam practice

1. How to Use This Guide

এই গাইড কীভাবে পড়বে

Step	What to do	বাংলা নির্দেশনা
1	Read the easy explanation	সহজ ব্যাখ্যাটি পড়ো
2	Learn the memory code	Memory code মুখস্থ করো
3	Close the book and recall	বই বন্ধ করে মনে করার চেষ্টা করো
4	Write a 5-mark answer	৫ নম্বরের উত্তর লিখে অনুশীলন করো
5	Take the MCQ and scenario test	MCQ ও scenario দিয়ে নিজেকে যাচাই করো

MEMORY CODE

R-W-R-T = Read -> Write -> Recall -> Test

পড়ো -> লেখো -> মনে করো -> পরীক্ষা দাও

EXAM TIP

For a 5-mark answer: Definition + 3 key points + short conclusion.

For a 10-mark answer: Definition + purpose + process + documents/examples + conclusion.

Use the exact lecture terms: design intent, commissioning agent, administrative closure, financial closure, lessons learned, project archive and sign-off.

Spaced revision plan: Day 1 -> Day 3 -> Day 7 -> the night before the exam.

Active recall (সক্রিয়ভাবে মনে করা): read a heading, close the guide, and write everything you remember without looking.

2. Master Map of the Lecture

পুরো লেকচারের সহজ মানচিত্র

Part	Main idea	Memory key
Commissioning	Turn design into an integrated, operating facility and verify performance.	D-C-O-S
Commissioning documents	Plan, specifications/manual, testing documents and acceptance standards.	PSD + STMTPC
Project closeout	Final administrative and logistical closure after acceptance.	IPS-FLAC-ST
Closing process	Record budget, schedule, completion, transitions and sign-off.	B-C-T
Staffing release	Plan next assignments, recognize staff, transfer roles and evaluate.	P-H-M-E
Financial closure	Finish transactions, close accounts, release assets/site and determine final cost.	N-C-F-D-V
Lessons & archive	Capture good/bad lessons and securely store project artifacts.	PPNR + SSDA
Contractor & customer closure	Remove access, transfer knowledge, review deliverables and obtain approval.	BAAK + RDOA
Templates	Checklist, closeout report, baselines, O&M, lessons, review dates and approvals.	C-R-B-O-L-A

MEMORY CODE

IPS-FLAC-ST = Introduction, Process, Staffing, Financial closure, Lessons, Archive, Contractor plan, Sign-off, Templates

Project closeout-এর ৯টি শিরোনাম একসাথে মনে রাখার কোড।

3. What Is Commissioning?

Commissioning = ভবন/সিস্টেমকে নকশা অনুযায়ী কার্যকরভাবে চালু ও যাচাই করার প্রক্রিয়া

Commissioning is a separate and distinct service. It may begin at the start of a project and continue until occupancy, and sometimes even after the owner occupies the building.

It is often provided as an additional service by an independent third party called the commissioning agent.

For large or complex projects, the commissioning agent manages and verifies the design performance of the building components and systems, especially those considered critical or necessary by the consultant and owner.

EASY BANGLA

Commissioning হলো নকশা, নির্মাণ, পরীক্ষা ও পরিচালনার মধ্যে সমন্বয় করে নিশ্চিত করা যে ভবনের সিস্টেমগুলো মালিকের প্রয়োজন অনুযায়ী কাজ করছে।

Occupancy = ভবন ব্যবহার বা দখল গ্রহণ।

Independent third party = চুক্তির প্রধান দুই পক্ষের বাইরে স্বাধীন তৃতীয় পক্ষ।

MEMORY CODE

S-T-O = Start of project -> Takeover/occupancy -> Sometimes after occupancy

Commissioning-এর সময়সীমা মনে রাখো: শুরু থেকে দখল, কখনও দখলের পরও।

EXAM TIP

Definition লিখে অবশ্যই timing এবং commissioning agent-এর কথা উল্লেখ করবে।

4. Commissioning as Quality Assurance

গুণমান নিশ্চিতকরণ প্রক্রিয়া

Commissioning includes activities that transform a facility design into a fully integrated and operating system.

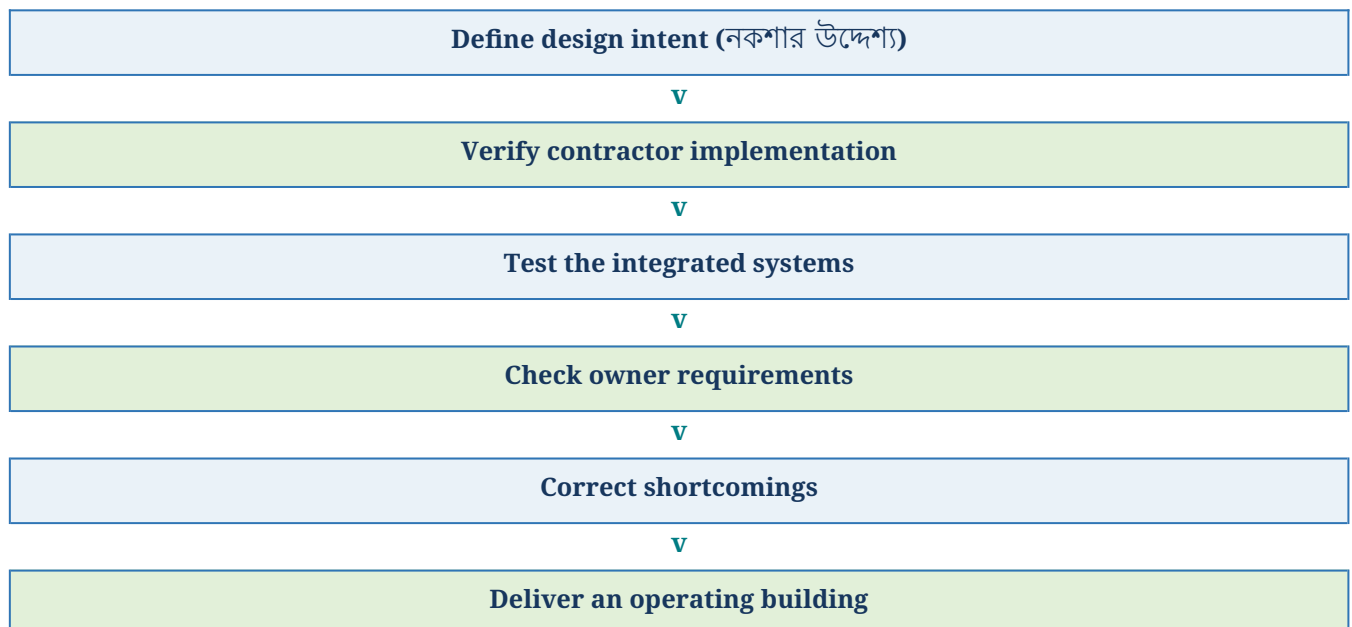
The lecture presents commissioning as a quality-assurance process that:

- Begins with the definition of the design intent and ends with delivery of the building.
- Confirms the contractor's implementation of the consultant's design as stated in the contract documents.
- Confirms that the consultant's design can satisfy the owner's defined requirements.
- Addresses any shortcomings discovered during verification and testing.

MEMORY CODE

D-C-O-S = Design intent -> Contractor implementation -> Owner requirements -> Shortcomings
নকশার উদ্দেশ্য -> ঠিকমতো বাস্তবায়ন -> মালিকের চাহিদা -> ত্রুটি সংশোধন

Commissioning QA flow



5. Role of the Commissioning Agent

Commissioning agent-এর ভূমিকা

Role	Easy explanation	বাংলা অর্থ
Independent verification	Checks whether design and construction satisfy owner objectives.	স্বাধীনভাবে নকশা ও নির্মাণ যাচাই
Contract performance	Verifies the contractor's performance of the contract.	ঠিকাদারের চুক্তি পালন যাচাই
Communication	Helps communicate design intent clearly to consultant and contractor.	নকশার উদ্দেশ্য স্পষ্টভাবে জানানো
Testing management	Explains and manages verification and testing procedures.	যাচাই ও পরীক্ষার পদ্ধতি পরিচালনা
Acceptance impact	Its tests and findings may affect acceptance of the work.	কাজ গ্রহণের সিদ্ধান্তে প্রভাব

MEMORY CODE

V-V-C-T-A = Verify design, Verify construction/contract, Communicate intent, Test, Affect acceptance
Agent-এর ৫ কাজ।

Early involvement is important because it improves communication of the design intent before construction decisions become fixed.

EXAM TIP

“Why should the commissioning agent be involved early?” প্রশ্নে communication, clear design intent এবং reduced misunderstanding লিখবে।

6. Commissioning Requirements in Bid Documents

দরপত্র নথিতে কমিশনিং তথ্য

The contractor usually joins the project team later. Therefore, verification and testing procedures should be explained in the bid documents.

This helps bidders evaluate:

- The time implication of commissioning activities.
- The cost implication of the commissioning agent's participation.
- How commissioning will affect acceptance of the work.

Bid documents prepared with commissioning-agent input should include:

Document	What it contains	বাংলা
Commissioning plan	Scope and sequence of the commissioning program.	কাজের পরিধি ও ধাপ
Commissioning specifications/manual	Verification forms, testing procedures and probable duration.	ফর্ম, পরীক্ষা পদ্ধতি ও সম্ভাব্য সময়
Specialized testing documents	Relevant testing standards or method options, such as the standards named in the lecture.	বিশেষ পরীক্ষা মানদণ্ড/পদ্ধতি

MEMORY CODE

P-S-D = Plan -> Specifications/manual -> Specialized documents
Bid documents-এর মূল ৩ অংশ।

7. Submission and Acceptance Standards

কোন কোন নথি/কার্যক্রম জমা ও গ্রহণ করতে হবে

The commissioning-related bid documents should set standards for submission and acceptance of the following:

Item	Purpose	বাংলা অর্থ
Shop drawings	Show detailed fabrication or installation information.	বিস্তারিত নির্মাণ/স্থাপন নকশা
Contractor's tests	Provide evidence that systems were tested.	ঠিকাদারের পরীক্ষা
Product, system, O&M manuals	Explain operation and maintenance requirements.	পণ্য/সিস্টেম পরিচালনা ও রক্ষণাবেক্ষণ নির্দেশিকা
Training programs	Prepare the owner's staff to operate the facility.	ব্যবহারকারী কর্মীদের প্রশিক্ষণ
Post-occupancy or seasonal testing	Verify performance after occupancy or under seasonal conditions.	দখলের পর/মৌসুমি পরীক্ষা
Pre-takeover activities	List activities to finish before owner takeover and occupancy.	মালিক গ্রহণের আগে সম্পন্ন কাজ

MEMORY CODE

S-T-M-T-P-C = Shop drawings, Tests, Manuals, Training, Post-occupancy testing, Commissioning activities
মনে রাখার বাক্য: "Shop Test Manual Training, Post-occupancy Complete."

8. Commissioning Summary and Common Confusions

গুরুত্বপূর্ণ পার্থক্য

Topic	Commissioning	Not the same as
Nature	A quality-assurance and verification process.	Only starting equipment for the first time.
Timing	May start at project beginning and continue after occupancy.	A single end-of-project event.
Focus	Integrated facility performance and owner requirements.	Only checking one item or one trade.
Person	May involve an independent commissioning agent.	The contractor alone deciding acceptance.
Result	Verified, operating systems and addressed shortcomings.	Automatic project closeout.

EXAM TIP

Commissioning এবং closeout এক নয়। Commissioning performance যাচাই করে; closeout প্রশাসনিক, আর্থিক ও logistical সমাপ্তি সম্পন্ন করে।
Commissioning may support acceptance, but customer sign-off and closeout documents are separate closeout tasks.

9. What Is Project Closeout?

Project closeout = প্রকল্পের আনুষ্ঠানিক সমাপনী ধাপ

The Project Closeout Phase is the last phase in the project lifecycle.

Closeout begins when the user accepts the project deliverables and the project oversight authority concludes that the project has met the established goals.

The major focus of closeout is administrative closure and logistics.

EASY BANGLA

Project deliverables = প্রকল্প থেকে গ্রাহককে দেওয়ার নির্ধারিত ফলাফল/নথি/পণ্য।

Administrative closure = চূড়ান্ত নথি, অনুমোদন, হিসাব, দায়িত্ব হস্তান্তর ও রেকর্ড সম্পন্ন করা।

Logistics = মানুষ, নথি, সম্পদ, প্রবেশাধিকার ও হস্তান্তরের ব্যবহারিক ব্যবস্থাপনা।

MEMORY CODE

A-G-A-L = Acceptance -> Goals met -> Administrative closure -> Logistics

Closeout শুরু ও focus মনে রাখার কোড।

10. Nine Areas of Project Closeout

লেকচারের ৯টি closeout area

No.	Area	Core question
1	Introduction	When and why does closeout begin?
2	Project closing process	What record, transitions and sign-off are needed?
3	Staffing transfer and release	Where will staff go and how are responsibilities transferred?
4	Financial closure	Are all transactions, accounts and assets closed?
5	Lessons learned	What should be repeated or avoided?
6	Project archive	What artifacts must be securely stored?
7	Contractor closing plan	Are permissions removed and knowledge transferred?
8	Customer project sign-off	Has the customer formally accepted and approved closure?
9	Closing process templates	Are checklist, questionnaire/report and transition controls complete?

MEMORY CODE

IPS-FLAC-ST = Introduction, Process, Staffing, Financial, Lessons, Archive, Contractor, Sign-off, Templates
৯টি শিরোনাম মনে রাখার master code।

11. Closeout Workflow from the Lecture Diagram

Slide 10-এর process diagram-এর সহজ ব্যাখ্যা

The process diagram on lecture page 10 shows closeout moving from project execution and user acceptance into contract closeout tasks, lessons learned, a transition checklist and a project closeout report. The project then moves into operations and support, followed by post-implementation review, a post-implementation report and a lessons-learned archive.

Simplified closeout-to-operations flow



MEMORY CODE

A-C-L-T-R-O-R-R-A = Accept -> Close tasks -> Learn -> Transition -> Report -> Operate -> Review -> Report -> Archive

Process diagram মনে রাখার chain।

12. Project Closing Process and Historical Record

ভবিষ্যৎ প্রকল্পের জন্য কী নথিভুক্ত করতে হবে

The project closing process creates a historical record of:

- Planned and actual budget and schedule.
- Recommendations for future projects of similar size and complexity.
- Staffing and skills used to meet objectives and goals.
- How customer expectations were managed.
- Lessons learned and official project sign-off.

Closeout report element	What to record
Project background	Background information that future team members may need to review.
Project completion	Whether objectives, benefits and deliverables were met to customer satisfaction.
Outstanding items	Items that must be addressed in future phases.

MEMORY CODE

B-S-S-C-L-S = Budget, Schedule, Staffing, Customer expectations, Lessons, Sign-off
Historical record-এর ৬ বিষয়।

13. Transition Information at Closeout

কী কী হস্তান্তর পরিকল্পনা লিখতে হবে

Transition area	Required information	বাংলা
Final budget vs actual	Document final budget, actual cost and relevant comments.	চূড়ান্ত বাজেট বনাম বাস্তব ব্যয়
Deliverable transition	Plan release of deliverables to the customer.	ফলাফল গ্রাহকের কাছে হস্তান্তর
Other resource transition	Identify non-personnel resources, release date and responsible person.	যন্ত্র/নথি/অন্যান্য সম্পদের হস্তান্তর
Technical transition	Support hours, rollout, issue escalation, documents, known questions and storage location.	প্রযুক্তিগত সহায়তা হস্তান্তর
Closure communication	Communicate project closure to stakeholders and interested parties.	সব পক্ষকে সমাপ্তির সংবাদ

MEMORY CODE

B-D-R-T-C = Budget, Deliverables, Resources, Technical support, Communication
Closeout transition-এর ৫ area।

EXAM TIP

“Technical transition” প্রশ্নে support hours, escalation, documentation location এবং known Q&A লিখবে।

14. Staffing Transfer and Release

কর্মীদের দায়িত্ব পরিবর্তন ও ছাড়

Staffing transfer/release is the process used to accommodate staff changes or shift project responsibilities as the project comes to a close.

Maintaining full staffing until completion is important. Employees are less likely to leave early when management actively manages their professional interests.

Process of releasing staff:

1. Plan the next assignment for project staff early; do not wait until the last minute.
2. Honor and thank staff and contractors for their contributions.
3. Consider current employees for ongoing maintenance and operations roles.
4. Give the resource manager performance-evaluation recommendations.

MEMORY CODE

P-H-M-E = Plan next work, Honor staff, Maintenance roles, Evaluation
Staff release-এর ৪ ধাপ।

EXAM TIP

শুধু “staff release” লিখবে না; full staffing until completion কেন গুরুত্বপূর্ণ তাও ব্যাখ্যা করবে।

15. Financial Closure

প্রকল্পের আর্থিক সমাপ্তি

Financial closure is the process of completing all project-related financial transactions, finalizing and closing project financial accounts, disposing of project assets and releasing the work site.

It is a prerequisite to project closure and post-implementation review.

It establishes final project costs so they can be compared with budgeted costs during the post review.

EASY BANGLA

Outstanding invoice = এখনো পরিশোধ বা নিষ্পত্তি হয়নি এমন বিল।

Financial account = প্রকল্পের আয়-ব্যয় ও লেনদেনের হিসাব।

Dispose of assets = সম্পদ নিয়ম অনুযায়ী বিক্রি, হস্তান্তর বা নিষ্পত্তি করা।

Release the work site = কাজের স্থান আনুষ্ঠানিকভাবে ছেড়ে দেওয়া।

MEMORY CODE

T-A-A-S-C = Transactions -> Accounts -> Assets -> Site -> Final Cost

Financial closure-এর সংজ্ঞা মনে রাখার code।

16. Process of Financial Closure

আর্থিক সমাপ্তির ধাপ

- 5. Verify that there are no outstanding invoices or unresolved financial obligations.
- 6. Close financial accounts under the applicable accounting procedures.
- 7. Comply with funding-source rules for records retention and financial reporting.
- 8. Set and communicate a deadline for all transactions, accounts and timesheets.
- 9. Verify that Statement of Work acceptance criteria have been met before paying vendors and consultants.

MEMORY CODE

N-C-F-D-V = No outstanding bills, Close accounts, Follow funding rules, Deadline, Verify acceptance
এ ধাপ এক লাইনে।

Check	Why it matters
Outstanding invoices	Prevents hidden unpaid obligations after closure.
Accounts closed	Creates final, reliable financial records.
Funding rules followed	Supports required reporting and retention.
Deadline published	Ensures staff and vendors complete transactions on time.
Acceptance before payment	Links final payment to required performance.

17. Lessons Learned

প্রকল্পের অভিজ্ঞতা থেকে শেখা

A lessons-learned session is valuable regardless of whether the project was successful.

Participants may include the project team, stakeholder representatives, management and, where applicable, the staff who will operate the product.

- Provide praise and recognition.
- Discuss improvements for future processes and procedures.
- Record both negative events and positive practices that helped the project.
- Avoid blame; identify ownership of problem areas and useful corrective recommendations.
- Submit lessons as they occur instead of waiting only for the post-implementation review.

MEMORY CODE

P-P-N-R = Praise, Positive & negative, No blame, Record early
Lessons learned-এর সঠিক attitude।

EXAM TIP

“Lessons learned” মানে শুধু ভুলের তালিকা নয়; ভালো কাজ পুনরাবৃত্তি ও খারাপ কাজ এড়ানোর পরিকল্পনা।

18. Project Archive

প্রকল্প নথি সংরক্ষণাগার

A project archive is the gathering and storing of project artifacts, such as the project charter and working documents, at project close.

Historical documents support future projects, ongoing operations and answers to later questions about the project.

Archiving process:

- 10. Use and document a clear folder structure for electronic artifacts.
- 11. Secure artifacts so they cannot be modified.
- 12. Describe the stored artifacts, software used, point of contact and folder structure.
- 13. Ensure the support team knows about and can access the artifacts.

MEMORY CODE
S-S-D-A = Structure, Secure, Describe, Access
Project archive-এর ৪ ধাপ।

Artifact example	Use after closeout
Project charter	Explains authority, purpose and original project basis.
Working documents	Supports technical or administrative questions.
Reports and baselines	Allows performance, schedule and cost review.
O&M and support documents	Supports ongoing operation and maintenance.

19. Contractor Closing Plan

ঠিকাদারের access বন্ধ ও knowledge transfer

After completion, a contractor closing plan must be executed. Its purpose is to remove permissions from the project environment and ensure that knowledge is transferred to the appropriate people.

Typical actions in the lecture:

- Return security badges.
- Disable access to applications.
- Disable application and database accounts.
- Submit an account-deletion form.
- Demonstrate knowledge transfer.

MEMORY CODE

B-A-A-K = Badges, Application access, Account deletion, Knowledge transfer
Contractor closing plan-এর ৪ key।

EXAM TIP

এটি শুধু “contract শেষ” নয়; security, permission এবং knowledge handover-এর কার্যকর পরিকল্পনা।

20. Customer Project Sign-off

গ্রাহকের আনুষ্ঠানিক গ্রহণ ও অনুমোদন

A key closeout task is customer acceptance of project deliverables.

The lecture recommends a final customer review of deliverables against requirements and specifications. Deviations should already be documented and approved.

During the meeting, participants review deviations and justifications, then work to close open action items and issues.

Bringing stakeholders together in one meeting avoids closing each open item separately.

Approval is certification by both the customer and project managers that the project has been officially closed.

MEMORY CODE

R-D-O-A = Review deliverables -> Deviations -> Open items -> Approval

Customer sign-off-এর ৪ ধাপ।

Term	বাংলা অর্থ
Requirements and specifications	চাহিদা ও কারিগরি শর্তাবলি
Deviation	নির্ধারিত মান থেকে বিচ্যুতি
Justification	বিচ্যুতির যৌক্তিক ব্যাখ্যা
Open item	এখনও বন্ধ হয়নি এমন কাজ/সমস্যা
Official sign-off	আনুষ্ঠানিক সমাপ্তির অনুমোদন

21. Project Closeout Checklist

Checklist কী যাচাই করে

The lecture identifies a separate closeout checklist covering project implementation and acceptance, final documentation, staffing plan and lessons learned.

The illustrated transition checklist on lecture pages 25-27 checks whether:

- Deliverables were accepted and any conditions were recorded.
- Performance goals were evaluated.
- Actual cost was compared with the approved cost baseline and changes were documented.
- Actual milestone dates were compared with the approved schedule and changes were documented.
- Scope changes and their effect on performance, cost and schedule were recorded.
- Operations management accepted responsibility, documents and training.
- Resources were transferred, documents archived, lessons recorded, and post-implementation review arrangements made.

MEMORY CODE

A-P-C-S-O-R-L-R = Acceptance, Performance, Cost, Schedule, Operations, Resources, Lessons, Review
Transition checklist-এর বড় headings।

22. Project Closeout Report: Main Sections

Slides 28-37-এর report forms সহজভাবে

Section	What the form records
A. General information	Project title, working title, sponsor/agency, preparer, date/control number.
B. Project deliverables	Deliverable, date accepted, contingencies or conditions.
C. Performance baseline	Business objective, performance goal and results.
D. Cost baseline	Planned, actual, variance and explanation for expenditure/funding categories.
E. Schedule baseline	Planned and actual dates, variance and explanation.
F. Scope	Scope changes and their impact.
G. Operations & maintenance	Transition plan and O&M cost information.
H. Project resources	Resource disposition, ongoing O&M need and responsible person.
I. Project documentation	Report/document, media, storage location and disposition.
J. Lessons learned	Problem, discussion, references and corrective actions.
K. Review/report dates	Post-implementation review/report dates and responsible person.
L. Approvals	Signatures/printed names, titles and dates.

MEMORY CODE

G-D-P-C-S-O-R-D-L-R-A = General, Deliverables, Performance, Cost, Schedule, Operations, Resources, Documentation, Lessons, Review, Approvals
Closeout report section chain!

23. Understanding Baselines and Variance

Performance, cost, schedule ও scope

Baseline	Meaning	What the closeout report compares
Performance baseline	Approved performance goals.	Goal versus actual result.
Cost baseline	Approved planned project cost/funding.	Planned versus actual cost and funding.
Schedule baseline	Approved project dates and milestones.	Planned versus actual completion dates.
Scope baseline	Approved work and deliverables.	Approved scope versus changes and their impact.

SIMPLE CALCULATION AID

Variance means the difference between the planned value and the actual value.
A common study convention is: $\text{Variance} = \text{Actual} - \text{Planned}$.
Always explain the reason for a major cost or schedule variance in the closeout report.

MEMORY CODE

P-C-S-S = Performance, Cost, Schedule, Scope
চারটি baseline/assessment area।

24. Operations, Resources, Documentation and Corrective Action

Closeout report-এর শেষ অংশগুলো

Area	Key content
Operations & maintenance	Describe transition to the organization that will operate and maintain the project result; compare operating costs where required.
Project resources	Identify each resource, whether it is needed for ongoing O&M, and who receives it.
Project documentation	Identify document/media, storage location and disposition.
Lessons learned	State the problem, discuss cause/impact, cite references and record corrective actions.
Post-implementation review	Set dates and assign a responsible person for review and report.
Approvals	Obtain signatures from the required project authorities.

MEMORY CODE

O-R-D-C-R-A = Operations, Resources, Documents, Corrective actions, Review, Approvals
Closeout report-এর final six areas।

25. Post-Implementation Report Format

Lecture page 38-এর report structure

The lecture’s post-implementation report format contains a cover page and a report body.

Part	Contents
Cover page	Title; organization/agency; date; author’s name; address; phone number; email address.
Opening sections	Executive summary; table of contents; introduction; background.
Method and project basis	Post-implementation review process or methodology; project proposal; project charter.
Results	Findings and recommendations.

MEMORY CODE
C-E-T-I-B-M-P-C-F-R = Cover, Executive summary, TOC, Introduction, Background, Method, Proposal, Charter, Findings, Recommendations
Post-implementation report-এর sequence।

EXAM TIP
PIR report নিয়ে প্রশ্ন এলে “findings” এবং “recommendations” দিয়ে শেষ করবে।

26. High-Value Comparison Tables

পরীক্ষায় পার্থক্য লিখতে ব্যবহার করো

Point	Commissioning	Project closeout
Main purpose	Verify integrated operation and design performance.	Complete administrative, financial and logistical closure.
Timing	May start at project beginning and continue after occupancy.	Begins after deliverable acceptance and goals are met.
Main actor	May include an independent commissioning agent.	Project team, customer, management and oversight authorities.
Key outputs	Verified tests, corrected shortcomings, operating systems.	Sign-off, financial closure, archive, lessons and transition.

Point	Financial closure	Customer sign-off
Focus	Money, accounts, assets and work site.	Acceptance of deliverables and official approval.
Key check	No unpaid obligations and acceptance before payment.	Requirements/specifications, deviations and open items.
Result	Final project cost and closed accounts.	Certification that the project is officially closed.

27. Essential Vocabulary with Bangla Meaning - Part 1

Commissioning and closeout terms

English	বাংলা অর্থ
Commissioning	কার্যকারিতা যাচাই ও চালু করার সমন্বিত প্রক্রিয়া
Commissioning agent	কমিশনিং প্রতিনিধি/স্বাধীন যাচাইকারী
Design intent	নকশার উদ্দেশ্য
Quality assurance	গুণমান নিশ্চিতকরণ
Integrated system	সমন্বিত সিস্টেম
Occupancy	ভবন ব্যবহার বা দখল গ্রহণ
Verification	যাচাই
Shortcoming	ত্রুটি বা ঘাটতি
Bid documents	দরপত্র নথি
Scope and sequence	কাজের পরিধি ও ধারাবাহিক ধাপ
Specification	কারিগরি শর্তাবলি
Testing procedure	পরীক্ষা পদ্ধতি
Shop drawing	বিস্তারিত নির্মাণ/স্থাপন নকশা
O&M manual	পরিচালনা ও রক্ষণাবেক্ষণ নির্দেশিকা
Seasonal testing	মৌসুমি পরীক্ষা
Takeover	দায়িত্ব বা দখল গ্রহণ
Project closeout	প্রকল্প সমাপনী কার্যক্রম
Deliverable	হস্তান্তরযোগ্য ফলাফল/পণ্য/নথি
Oversight authority	তদারকি কর্তৃপক্ষ
Administrative closure	প্রশাসনিক সমাপ্তি
Logistics	ব্যবহারিক সম্পদ ও কার্যব্যবস্থাপনা
Historical record	ঐতিহাসিক নথি
Official sign-off	আনুষ্ঠানিক অনুমোদন
Outstanding item	অসমাপ্ত বিষয়

28. Essential Vocabulary with Bangla Meaning - Part 2

Financial, transition and reporting terms

English	বাংলা অর্থ
Transition	হস্তান্তর/রূপান্তর
Stakeholder	স্বার্থসংশ্লিষ্ট পক্ষ
Escalation	উচ্চতর পর্যায়ে বিষয় উত্থাপন
Staff release	কর্মী ছাড়/দায়িত্ব থেকে অবমুক্তি
Financial closure	আর্থিক সমাপ্তি
Outstanding invoice	বকেয়া বিল
Financial obligation	আর্থিক দায়
Records retention	নথি সংরক্ষণ
Acceptance criteria	গ্রহণযোগ্যতার মানদণ্ড
Statement of Work	কাজের বিবরণ/চুক্তির কাজের পরিধি
Lessons learned	অভিজ্ঞতা থেকে শেখা বিষয়
Corrective action	সংশোধনমূলক ব্যবস্থা
Project archive	প্রকল্প সংরক্ষণাগার
Artifact	প্রকল্প নথি বা উৎপাদিত উপাদান
Knowledge transfer	জ্ঞান হস্তান্তর
Security badge	প্রবেশাধিকার পরিচয়পত্র
Deviation	বিচ্যুতি
Justification	যৌক্তিকতা
Baseline	অনুমোদিত ভিত্তিরেখা
Variance	পার্থক্য/বিচ্যুতি
Post-implementation review	বাস্তবায়ন-পরবর্তী পর্যালোচনা
Executive summary	নির্বাহী সারাংশ
Project charter	প্রকল্প সনদ
Findings	পর্যবেক্ষণ/ফলাফল
Recommendations	সুপারিশ

29. Model Written Answers

পরীক্ষায় উত্তর লেখার নমুনা

Model 1: Define commissioning and explain its purpose. (5 marks)

Commissioning is a separate quality-assurance service that may start at the beginning of a project and continue until or after occupancy. Its purpose is to transform the facility design into a fully integrated operating system. It confirms the contractor's implementation of the consultant's design, checks whether the design satisfies the owner's requirements, and addresses shortcomings. Large or complex projects may use an independent commissioning agent to manage and verify this process.

Model 2: Explain project closeout. (5 marks)

Project closeout is the last phase of the project lifecycle. It begins after the user accepts the deliverables and the oversight authority determines that project goals have been met. Its main focus is administrative closure and logistics. Typical closeout work includes the closing process, staff release, financial closure, lessons learned, project archive, contractor access closure, customer sign-off and final templates/reports.

Model 3: Describe the process of financial closure. (5 marks)

Financial closure completes all project financial transactions and closes the accounts. The project team verifies that no invoices or obligations remain, follows accounting and funding rules, sets a deadline for transactions and timesheets, and confirms Statement of Work acceptance before paying vendors and consultants. The result is a reliable final project cost for comparison with the budget.

30. Model Written Answers - Continued

Long-answer structure

Model 4: Discuss lessons learned and project archive. (10 marks)

A lessons-learned session includes the project team, stakeholders, management and sometimes operations staff. It records both positive and negative experiences, recognizes contributions, avoids blame and produces useful corrective recommendations. Lessons should be submitted as they occur. A project archive then gathers and stores important artifacts. The archive should use a documented folder structure, protect files from alteration, describe the contents and software, identify a contact person, and provide support-team access. Together, lessons and archives improve future projects and ongoing support.

Model 5: Explain customer sign-off and contractor closing plan. (10 marks)

Customer sign-off is the formal acceptance of deliverables. A final customer review compares deliverables with requirements and specifications, discusses approved deviations and closes open items. The customer and project managers then certify official closure. Separately, the contractor closing plan removes security badges and application/database access, submits account-deletion requests and demonstrates knowledge transfer. These two processes protect both acceptance and operational security.

EXAM TIP

Long answer-এর structure: Definition -> Purpose -> Steps -> Documents/Examples -> Conclusion.
প্রতিটি answer-এ lecture-এর memory code ব্যবহার করলে points বাদ পড়বে না।

31. Probable Written Questions

সম্ভাব্য পরীক্ষার প্রশ্ন

14. Define commissioning. How is it different from a single testing event?
15. Explain commissioning as a quality-assurance process.
16. Discuss the role and early involvement of a commissioning agent.
17. What commissioning information should be included in bid documents?
18. List the standards for submission and acceptance of commissioning documents and activities.
19. Define project closeout and explain when it begins.
20. Describe the nine areas of project closeout.
21. Explain the project closing process and its historical-record purpose.
22. Discuss deliverable, resource, technical and communication transitions.
23. Explain staffing transfer and release.
24. Define financial closure and describe its process.
25. Explain the purpose and correct approach of a lessons-learned session.
26. What is a project archive? Describe the archiving process.
27. Explain the contractor closing plan.
28. Describe the customer project sign-off process.
29. What does the project closeout transition checklist verify?
30. Describe the major sections of a project closeout report.
31. Compare performance, cost, schedule and scope baselines.
32. Explain the post-implementation report format.
33. Differentiate commissioning, project closeout and financial closure.

32. MCQ Practice - 20 Questions

প্রথমে উত্তর না দেখে চেষ্টা করো

1. Commissioning may begin:

- A. Only after handover
- B. At the beginning of the project
- C. Only after final payment
- D. Only after occupancy

2. A commissioning agent is often:

- A. An independent third party
- B. Only the owner's accountant
- C. Only the supplier
- D. The final customer

3. Commissioning quality assurance begins with:

- A. Final payment
- B. Design intent
- C. Staff release
- D. Project archive

4. The D-C-O-S code includes:

- A. Owner requirements
- B. Final invoices
- C. Security badges
- D. Project sponsorship

5. Bid documents should explain commissioning to help bidders evaluate:

- A. Only design colour
- B. Time and cost implications
- C. Only staff salary
- D. Tax law

6. Which is part of commissioning submission standards?

- A. Shop drawings
- B. Project sponsor signature only
- C. Final payroll only
- D. Archive destruction

7. Project closeout is:

- A. The first project phase
- B. The last project phase
- C. A design-only task
- D. A procurement method

8. Closeout begins after:

- A. User acceptance and goals met
- B. First bid submission
- C. Design concept only
- D. Staff recruitment

9. The main focus of closeout is:

- A. Administrative closure and logistics
- B. Only structural design
- C. Only marketing
- D. Only site excavation

10. Financial closure is a prerequisite to:

- A. Project closure and post-implementation review
- B. Initial bidding only

- C. Design intent
- D. Shop drawing preparation

11. Before vendor payment, the team should verify:

- A. Acceptance criteria in the Statement of Work
- B. Only employee attendance
- C. Only archive folder names
- D. Only security badges

12. Lessons learned should record:

- A. Only failures
- B. Only successes
- C. Both positive and negative experience
- D. No project experience

13. A good lessons-learned discussion should:

- A. Assign blame
- B. Avoid blame and develop recommendations
- C. Delete records
- D. Ignore positive practices

14. Project archive artifacts should be:

- A. Open to unauthorized modification
- B. Secured against modification
- C. Immediately destroyed
- D. Stored without description

15. BAAK is the code for:

- A. Contractor closing actions
- B. Financial closure
- C. Commissioning documents
- D. Customer sign-off

16. Customer sign-off includes:

- A. Reviewing deliverables against requirements
- B. Skipping open items
- C. Avoiding stakeholder meetings
- D. Deleting deviations

17. RDOA ends with:

- A. Archive
- B. Approval
- C. Accounting
- D. Occupancy

18. The cost baseline form compares:

- A. Planned and actual cost
- B. Only actual staff names
- C. Only design intent
- D. Only customer comments

19. A post-implementation report ends with:

- A. Findings and recommendations
- B. Shop drawings
- C. Security badges
- D. Bid bonds

20. The page-10 workflow moves from closeout toward:

- A. Operations and support
- B. Initial tender advertising
- C. Land acquisition only
- D. Concept design only

ANSWER KEY

1-B, 2-A, 3-B, 4-A, 5-B, 6-A, 7-B, 8-A, 9-A, 10-A, 11-A, 12-C, 13-B, 14-B, 15-A, 16-A, 17-B, 18-A, 19-A, 20-A

33. Scenario-Based Practice

ঘটনা দেখে concept চিনে উত্তর দাও

Scenario 1: A complex building is almost complete, but its systems do not work together as intended. Which process and person are relevant?

Suggested answer

Commissioning; an independent commissioning agent may verify integrated performance, owner requirements and shortcomings.

Scenario 2: A contractor argues that testing requirements were not included in its price. What document issue should have been managed?

Suggested answer

Verification/testing procedures, scope, sequence and probable duration should have been clearly included in bid documents.

Scenario 3: A project is accepted, but unpaid invoices and open accounts remain. Can closeout be considered complete?

Suggested answer

No. Financial closure is a prerequisite and requires resolving invoices, obligations and accounts.

Scenario 4: A team meeting discusses only failures and blames one employee. What is wrong?

Suggested answer

Lessons learned should include positive and negative points, avoid blame and create useful recommendations.

Scenario 5: The contractor has finished but still has system access and has not transferred operating knowledge.

Suggested answer

The contractor closing plan is incomplete; badges/access/accounts must be removed and knowledge transfer demonstrated.

Scenario 6: The customer accepts most work but several deviations are undocumented.

Suggested answer

Final sign-off should wait until deviations are documented, approved and reviewed with open items.

34. Active Recall Quiz

বই বন্ধ করে উত্তর দাও

34. When can commissioning start and end?
35. Who is a commissioning agent?
36. What does D-C-O-S stand for?
37. Why should commissioning procedures appear in bid documents?
38. What does P-S-D stand for?
39. List the six submission/acceptance items in STMTPC.
40. When does project closeout begin?
41. What are the two main closeout focuses?
42. State the nine closeout areas in IPS-FLAC-ST.
43. Describe the page-10 closeout workflow.
44. What six things belong in the historical record?
45. What does BDRTC stand for?
46. Why maintain full project staffing until completion?
47. State the four staff-release steps.
48. Define financial closure.
49. State the NCFDV financial-closeout steps.
50. Who attends a lessons-learned session?
51. What does PPNR mean?
52. What does SSDA mean for archiving?
53. What does BAAK mean for contractor closing?
54. What does RDOA mean for customer sign-off?
55. What does the transition checklist verify?
56. Name the major closeout report sections.
57. Differentiate the four baselines.
58. What are the parts of the post-implementation report?

EXAM TIP

যে প্রশ্নের উত্তর ৩০ সেকেন্ডে বলতে পারো না, সেই section আবার পড়ো।

35. Seven-Day Revision Plan

পরীক্ষার আগের ৭ দিনের পরিকল্পনা

Day	Focus	Task
Day 1	Commissioning basics	Learn definition, timing, agent and D-C-O-S.
Day 2	Bid documents	Learn P-S-D and STMTPC; write one 5-mark answer.
Day 3	Closeout overview	Learn AGAL, IPS-FLAC-ST and the workflow diagram.
Day 4	People and finance	Study staff release and N-C-F-D-V financial closure.
Day 5	Knowledge and records	Study lessons learned, archive and contractor closing.
Day 6	Sign-off and reports	Study RDOA, checklist, baselines and report sections.
Day 7	Mock exam	Answer 5 written questions, 20 MCQs and all active-recall prompts.

NIGHT-BEFORE RULE

Revise memory codes and comparison tables only.

Do not start a new topic.

Write the nine closeout areas and the five financial-closeout steps once from memory.

36. One-Page Master Revision Sheet

শেষ মুহূর্তের revision

Topic	Must remember
Commissioning	Separate QA service; begins early; may continue after occupancy; independent agent.
QA code	D-C-O-S: design intent, contractor implementation, owner requirements, shortcomings.
Bid documents	P-S-D: plan, specifications/manual, specialized testing documents.
Acceptance items	STMTPC: shop drawings, tests, manuals, training, post-occupancy, commissioning activities.
Closeout start/focus	User accepts deliverables + goals met; administrative closure + logistics.
Nine areas	IPS-FLAC-ST.
Transitions	BDRTC: budget, deliverables, resources, technical, communication.
Staff release	P-H-M-E.
Financial closure	N-C-F-D-V.
Lessons learned	P-P-N-R.
Archive	S-S-D-A.
Contractor close	B-A-A-K.
Customer sign-off	R-D-O-A.
Baselines	Performance, cost, schedule and scope.
Post-review report	Cover + executive summary + TOC + introduction/background + method/proposal/chapter + findings/recommendations.

FINAL ANSWER FORMULA

D-P-P-D-C = Definition -> Purpose -> Process -> Documents/details -> Conclusion.

Use headings and underline memory-code keywords in the exam.

Always write both the technical term and a simple explanation.

Best wishes for your term final examination

শান্তভাবে পড়ো, **active recall** করো, আর **structured answer** লেখো।